

Instructions: All instructions remain the same from our previous assessments.

Derivatives

$$\frac{d}{dx}(\sin^{-1} x) = \frac{1}{\sqrt{1-x^2}}$$

$$\frac{d}{dx}(\cos^{-1} x) = -\frac{1}{\sqrt{1-x^2}}$$

$$\frac{d}{dx}(\tan^{-1} x) = \frac{1}{1+x^2}$$

$$\frac{d}{dx}(\sinh^{-1} x) = \frac{1}{\sqrt{1+x^2}}$$

$$\frac{d}{dx}(\cosh^{-1} x) = \frac{1}{\sqrt{x^2-1}}$$

$$\frac{d}{dx}(\tanh^{-1} x) = \frac{1}{1-x^2}$$

Moments and Centroid

$$M_x = \frac{\rho}{2} \int_a^b (f(x)^2 - g(x)^2) dx$$

$$M_y = \rho \int_a^b x(f(x) - g(x)) dx$$

$$m = \rho \int_a^b (f(x) - g(x)) dx$$

$$(\bar{x}, \bar{y}) = \left(\frac{M_y}{m}, \frac{M_x}{m} \right)$$

Integrals

$$\int u dv = uv - \int v du$$

$$\int \tan x dx = \ln |\sec x| + C$$

$$\int \sec x dx = \ln |\sec x + \tan x| + C$$

$$\int \frac{dx}{x^2 + a^2} = \frac{1}{a} \tan^{-1} \left(\frac{x}{a} \right) + C$$

Polar Coordinates

$$\frac{dy}{dx} = \frac{\frac{dr}{d\theta} \sin \theta + r \cos \theta}{\frac{dr}{d\theta} \cos \theta - r \sin \theta}$$

$$A = \int_a^b \frac{1}{2} (f(\theta)^2 - g(\theta)^2) d\theta$$

$$L = \int_a^b \sqrt{r^2 + \left(\frac{dr}{d\theta} \right)^2} d\theta$$

Surface Area

$$A = \int_a^b 2\pi f(x) \sqrt{1 + f'(x)^2} dx$$

$$A = \int_a^b 2\pi x \sqrt{1 + f'(x)^2} dx$$

Area Between Curves

$$A = \int_a^b (f(x) - g(x)) dx$$

Work

$$W = \int_a^b F(x) dx$$

Hydrostatic Force

$$F = \rho g \int_a^b w(y) d(y) dy$$

Arc Length

$$L = \int_a^b \sqrt{1 + f'(x)^2} dx$$

Approximate Integration

$$|E_T| \leq \frac{K(b-a)^3}{12n^2}$$

$$|E_M| \leq \frac{K(b-a)^3}{24n^2}$$

$$|E_S| \leq \frac{L(b-a)^5}{180n^4}$$

Volumes

$$\text{Cross sections: } V = \int_a^b A(x) dx$$

$$\text{Cylindrical shells: } V = \int_a^b 2\pi x f(x) dx$$

Hyperbolic Functions

$$\cosh x = \frac{e^x + e^{-x}}{2}$$

$$\sinh x = \frac{e^x - e^{-x}}{2}$$

$$\tanh x = \frac{\sinh x}{\cosh x}$$

$$\sinh^{-1} x = \ln(x + \sqrt{x^2 + 1})$$

$$\cosh^{-1} x = \ln(x + \sqrt{x^2 - 1})$$

$$\tanh^{-1} x = \frac{1}{2} \ln \left(\frac{1+x}{1-x} \right)$$

Trig & Hyperbolic Identities

$$\cos^2 x + \sin^2 x = 1$$

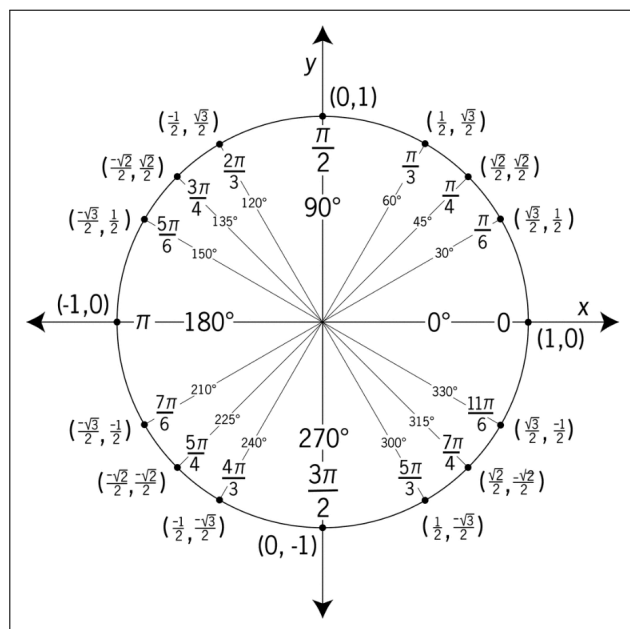
$$\tan^2 x + 1 = \sec^2 x$$

$$\sin^2 x = \frac{1}{2}(1 - \cos 2x)$$

$$\cos^2 x = \frac{1}{2}(1 + \cos 2x)$$

$$\cosh^2 x - \sinh^2 x = 1$$

$$1 - \tanh^2 x = \operatorname{sech}^2 x$$

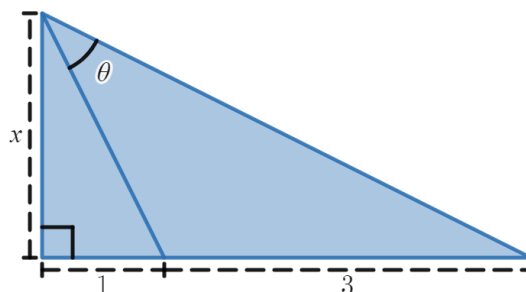


A1 - Inverse trigonometric and hyperbolic functions (inspired by Brightspace problem 4 and part 2 of the class notes)

(a) Consider the right triangle shown on the right.

- (i) Express θ as a function of x . *Hint: θ is the difference of two angles whose tangents can be expressed using information from the diagram.*

$$\theta = \tan^{-1}\left(\frac{y}{x}\right) - \tan^{-1}\left(\frac{1}{x}\right)$$



(b) Differentiate $f(x) = \arccos \ln(x)$.

$$f'(x) = \frac{-1}{\sqrt{1 - \ln(x)^2}} \cdot \frac{1}{x}$$

(c) Recall that $\sinh(x) = \frac{e^x - e^{-x}}{2}$.

- (i) Show that $\sinh^{-1}(x) = \ln(x + \sqrt{x^2 + 1})$ by solving for y in the equation $x = \frac{e^y - e^{-y}}{2}$.

$$2x = e^y - e^{-y}$$

$$2xe^y = e^{2y} - 1$$

$$e^{2y} - 2xe^y - 1 = 0$$

$$e^y = \frac{2x \pm \sqrt{4x^2 + 4}}{2}$$

$$e^y = x \pm \sqrt{x^2 + 1}$$

must take + as $e^y > 0$

$$e^y = x + \sqrt{x^2 + 1}$$

$$y = \ln(x + \sqrt{x^2 + 1}) \quad \checkmark$$

- (ii) Use implicit differentiation with the equation $\sinh(y) = x$ to show $\frac{d}{dx}(\sinh^{-1}(x)) = \frac{1}{\sqrt{1+x^2}}$.

$$\cosh(y) \cdot \frac{dy}{dx} = 1$$

$$\frac{dy}{dx} = \frac{1}{\cosh(y)} = \frac{1}{\sqrt{1 + (\sinh y)^2}} = \frac{1}{\sqrt{1 + x^2}} \quad \checkmark$$

A2 - L'Hospital's rule (inspired by Brightspace problems 2 and 10)

Evaluate each of the following limits.

$$(a) \lim_{x \rightarrow 0} (1 + x^2 + x \sin(x))^{\frac{1}{x^2}} = L$$

$$\ln(L) = \lim_{x \rightarrow 0} \frac{\ln(1 + x^2 + x \sin(x))}{x^2} \stackrel{H}{=} \lim_{x \rightarrow 0} \frac{\frac{2x + \sin(x) + x \cos(x)}{1 + x^2 + x \sin(x)}}{2x} = \lim_{x \rightarrow 0} \frac{2x + \sin(x) + x \cos(x)}{2x + 2x^3 + 2x^2 \sin(x)}$$

$$\stackrel{H}{=} \lim_{x \rightarrow 0} \frac{2 + \cos(x) + \cos(x) - x \sin(x)}{2 + 6x^2 + 4x \sin(x) + 2x^2 \cos(x)}$$

$$= \frac{4}{2} = 2$$

$$\rightarrow \boxed{L = e^2}$$

$$(b) \lim_{x \rightarrow 0} x^{-4} \cdot \left(\cos(x) - 1 + \frac{x^2}{2} \right)$$

$$\stackrel{H}{=} \lim_{x \rightarrow 0} \frac{-\sin(x) + x}{4x^3}$$

$$\stackrel{H}{=} \lim_{x \rightarrow 0} \frac{-\cos(x) + 1}{12x^2}$$

$$\stackrel{H}{=} \lim_{x \rightarrow 0} \frac{\sin(x)}{24x}$$

$$\stackrel{H}{=} \lim_{x \rightarrow 0} \frac{\cos(x)}{24} = \boxed{\frac{1}{24}}$$

A3 - Integration techniques I (inspired by part 1 of the class notes and past final worksheet problem 3)

Evaluate each of the following integrals. You do not need to simplify your answers.

(a) $\int_0^{0.5} \cos^{-1}(x) dx$

$$u = \cos^{-1}(x)$$

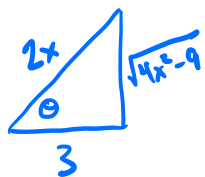
$$v = x$$

$$du = \frac{-1}{\sqrt{1-x^2}} dx \quad dv = dx$$

$$= x \cos^{-1}(x) \Big|_0^{0.5} + \int_0^{0.5} \frac{x}{\sqrt{1-x^2}} dx$$

$$= \frac{\pi}{6} - \sqrt{1-x^2} \Big|_0^{0.5} = \frac{\pi}{6} - (\sqrt{1-0.25} - \sqrt{1})$$

(b) $\int \frac{dx}{(4x^2 - 9)^{\frac{3}{2}}}$



$$x = \frac{3}{2} \sec \theta$$

$$dx = \frac{3}{2} \sec \theta \tan \theta d\theta$$

$$u = \sin \theta$$

$$\int \frac{\frac{3}{2} \sec \theta \tan \theta d\theta}{(9 \tan^2 \theta)^{\frac{3}{2}}} = \frac{1}{18} \int \frac{\sec \theta}{\tan^3 \theta} d\theta = \frac{1}{18} \int \frac{\cos \theta}{\sin^3 \theta} d\theta = \frac{1}{18} \int \frac{1}{u^2} du$$

$$= \frac{-1}{18u} + C = \frac{-1}{18 \sin \theta} + C$$

$$= \frac{-1}{18} \csc \theta + C = \boxed{\frac{-x}{9\sqrt{4x^2-9}} + C}$$

A4 - Integration techniques II (inspired by part 1 of the class notes and past final worksheet problem 1)

For each of the following integrals, evaluate it or show that it is divergent.

$$(a) \int \frac{x^5 + x^4 - 1}{x^4 + x^2} dx = \int \left(x+1 - \frac{x^3+x^2+1}{x^4+x^2} \right) dx \quad \frac{x^3+x^2+1}{x^2(x^2+1)} = \frac{A}{x} + \frac{B}{x^2} + \frac{Cx+D}{x^2+1}$$

$$\begin{array}{r} x^4+x^2 \overline{) x^5+x^4+0x^3+0x^2+0x-1} \\ \underline{-x^5 \quad +x^3} \\ 0+x^4-x^3 \\ \underline{-x^4 \quad +x^2} \\ -x^3-x^2-1 \end{array}$$

$$x^3+x^2+1 = Ax(x^2+1) + B(x^2+1) + (Cx+D)x^2$$

$$x=0 \rightarrow 1=B$$

$$x^3: 1=A+C$$

$$x^2: 1=B+D \rightarrow D=0$$

$$x: 0=A \quad (\text{so from above } C=1)$$

$$\int \left(x+1 - \frac{1}{x^2} - \frac{x}{x^2+1} \right) dx = \boxed{\frac{x^2}{2} + x + \frac{1}{x} - \frac{1}{2} \ln(x^2+1) + C}$$

$$(b) \int_{-\infty}^{\infty} \frac{e^x}{e^{2x}+3} dx$$

$$u=e^x$$

$$du=e^x dx$$

$$\int \frac{1}{u^2+3} du = \frac{1}{\sqrt{3}} \arctan\left(\frac{u}{\sqrt{3}}\right) = \frac{1}{\sqrt{3}} \arctan\left(\frac{e^x}{\sqrt{3}}\right)$$

$$\lim_{x \rightarrow \infty} \frac{1}{\sqrt{3}} \arctan\left(\frac{e^x}{\sqrt{3}}\right) = \frac{1}{\sqrt{3}} \cdot \frac{\pi}{2}$$

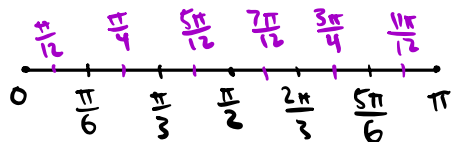
$$\lim_{x \rightarrow -\infty} \frac{1}{\sqrt{3}} \arctan\left(\frac{e^x}{\sqrt{3}}\right) = 0$$

$$\rightarrow \boxed{\frac{\pi}{2\sqrt{3}}}$$

A5 - Approximate integration (inspired by past final worksheet problem 5 and the class notes)

Consider the integral $I = \int_0^\pi \cos(2x)dx$, which we know is equal to 0 by straightforward antidifferentiation.

- (a) Use the midpoint rule with $n = 6$ to approximate I . You don't need to simplify your answer.



$$\frac{\pi}{6} \cdot \left(\cos\left(2 \cdot \frac{\pi}{12}\right) + \cos\left(2 \cdot \frac{\pi}{4}\right) + \cos\left(2 \cdot \frac{5\pi}{12}\right) + \cos\left(2 \cdot \frac{7\pi}{12}\right) + \cos\left(2 \cdot \frac{3\pi}{4}\right) + \cos\left(2 \cdot \frac{11\pi}{12}\right) \right)$$

- (b) Find an upper bound on the error of your approximation from part (a) using techniques covered in class. You don't need to simplify your answer.

$$|E_M| \leq \frac{K(b-a)^3}{24n^2}$$

$f'(x) = -2\sin(2x)$
 $f''(x) = -4\cos(2x)$

$|f''(x)| \leq 4$ on $[0, \pi]$ so $K=4$

$$|E_M| \leq \frac{4 \cdot (\pi - 0)^3}{24 \cdot 6^2}$$

- (c) Use Simpson's rule with $n = 6$ to approximate I . You don't need to simplify your answer.

$$\frac{\frac{\pi}{6}}{3} \left(\cos(2 \cdot 0) + 4 \cdot \cos\left(2 \cdot \frac{\pi}{6}\right) + 2 \cos\left(2 \cdot \frac{\pi}{3}\right) + 4 \cos\left(2 \cdot \frac{\pi}{2}\right) + 2 \cos\left(2 \cdot \frac{2\pi}{3}\right) + 4 \cos\left(2 \cdot \frac{5\pi}{6}\right) + \cos(2 \cdot \pi) \right)$$

- (d) How large should n be to guarantee that your approximation from part (c) is accurate to within 0.00001. You don't need to simplify your answer.

$$|E_s| \leq \frac{L \cdot (b-a)^5}{180n^4} \leq 0.00001$$

$$f'''(x) = 8\sin(2x)$$

$$f^{(4)}(x) = 16\cos(2x)$$

$\rightarrow L=16$

$$\frac{16 \cdot (\pi - 0)^5}{180 \cdot n^4} \leq 0.00001$$

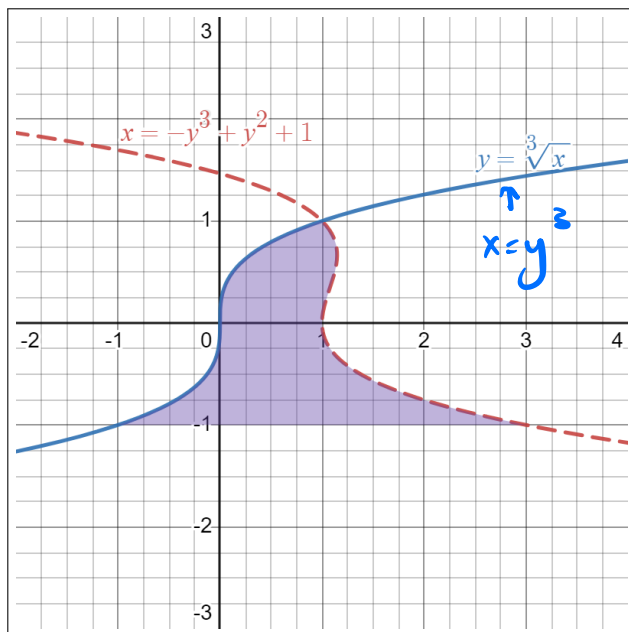
$$\rightarrow n \geq \sqrt[4]{\frac{16\pi^5}{180 \cdot 0.00001}}$$

B1 - Areas between curves (inspired by Brightspace problem 10 and the class notes)

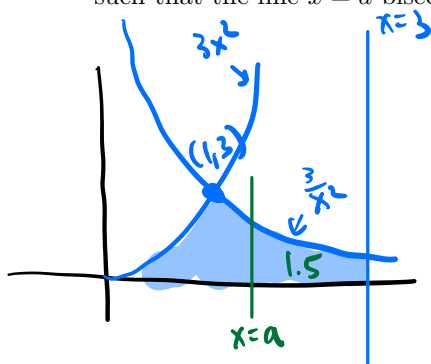
- (a) Setup, but do not simplify or evaluate, an integral representing the area of the shaded region to the right. The dotted curve is $x = -y^3 + y^2 + 1$ and the solid curve is $y = \sqrt[3]{x}$.

Curves intersect at (1,1)

$$\int_{-1}^1 (-y^3 + y^2 + 1 - y^3) dy$$



- (b) Consider the area of the region bounded by $y = 3x^2$, $y = \frac{3}{x^2}$, $y = 0$, and $x = 3$. Find the number a such that the line $x = a$ bisects this area.



$$3x^2 = \frac{3}{x^2} \rightarrow x = 1$$

$$\text{Total area: } \int_0^1 3x^2 dx + \int_1^3 \frac{3}{x^2} dx = x^3 \Big|_0^1 + \frac{-3}{x} \Big|_1^3 = 1 + 2 = 3$$

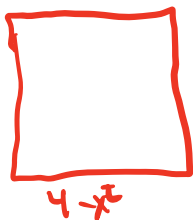
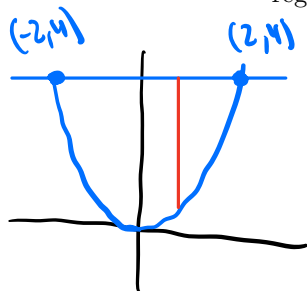
$$\frac{3}{2} = 1.5 > \int_0^1 3x^2 dx, \text{ so } a > 1.$$

$$1.5 = \int_a^3 \frac{3}{x^2} dx = \frac{-3}{x} \Big|_a^3 = \frac{-3}{3} - \frac{-3}{a} \rightarrow 2.5 = \frac{3}{a} \rightarrow a = \frac{3}{2.5} = \frac{6}{5}$$

B2 - Volumes (inspired by Brightspace problem 5 and part 2 of the class notes)

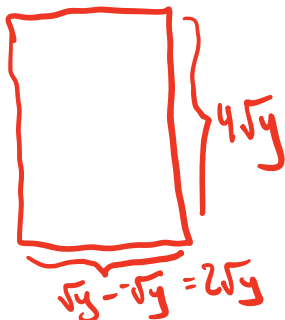
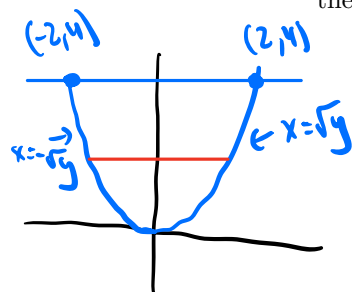
(a) Consider the region enclosed by the parabola $y = x^2$ and the line $y = 4$.

(i) Setup, but do not evaluate, an integral representing the volume of the solid whose base is this region and whose cross sections perpendicular to the x -axis are squares.



$$V = \int_{-2}^2 (4 - x^2)^2 dx$$

(ii) Setup, but do not evaluate, an integral representing the volume of the solid whose base is this region and whose cross sections perpendicular to the y -axis are rectangles with height two times their base.



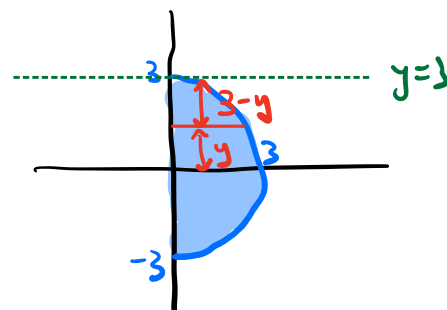
$$V = \int_0^4 2\sqrt{y} \cdot 4\sqrt{y} dy$$

(b) Setup, but do not evaluate, an integral representing the volume of the solid obtained by rotating the region bound by $x = \sqrt{9 - y^2}$ and the y -axis about each of the following lines:

(i) $y = 3$

Shells:

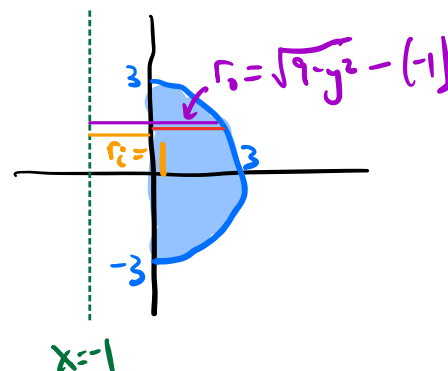
$$V = \int_{-3}^3 2\pi (3 - y) \cdot \sqrt{9 - y^2} dy$$



(ii) $x = -1$

cross sections:

$$V = \int_{-3}^3 (\pi (\sqrt{9 - y^2} + 1)^2 - \pi (1)^2) dy$$



B3 - Arclength and surface area (inspired by past final worksheet problem 2 and Brightspace problem 4)

(a) Let $f(x) = \left(4 - x^{\frac{2}{3}}\right)^{\frac{3}{2}}$.

(i) Find the arclength function for $f(x)$ starting from $x = 0$.

$$s(x) = \int_0^x \sqrt{1 + f'(t)^2} dt \quad f'(t) = \frac{3}{2} \left(4 - t^{\frac{2}{3}}\right)^{\frac{1}{2}} \cdot \left(-\frac{2}{3} t^{-\frac{1}{3}}\right)$$

$$f'(t)^2 + 1 = \frac{9}{4} \left(4 - t^{\frac{2}{3}}\right) \cdot \left(\frac{4}{9} \cdot t^{-\frac{2}{3}}\right) + 1 = 4t^{-\frac{2}{3}} - 1 + 1 = 4t^{-\frac{2}{3}}$$

$$= \int_0^x \sqrt{4t^{-\frac{2}{3}}} dt = \int_0^x 2t^{-\frac{1}{3}} dt = \boxed{3x^{\frac{2}{3}}}$$

(ii) Use the arclength function to find a number b such that the length of $f(x)$ from $x = 0$ to $x = b$ is 6.

$$3b^{\frac{2}{3}} = 6 \rightarrow b^{\frac{2}{3}} = 2 \rightarrow \boxed{b = \sqrt[3]{8}}$$

(b) Find the surface area of the shape obtained by rotating the curve $y = \sqrt{1 + 2e^x}$, $0 \leq x \leq 1$, about the x -axis.

$$y' = \frac{1}{\cancel{2}\sqrt{1+2e^x}} \cdot \cancel{2}e^x \quad (y')^2 + 1 = \frac{e^{2x}}{1+2e^x} + 1 = \frac{e^{2x}}{1+2e^x} + \frac{1+2e^x}{1+2e^x} = \frac{e^{2x} + 2e^x + 1}{1+2e^x} = \frac{(1+e^x)^2}{1+2e^x}$$

$$S = \int_0^1 2\pi \cdot \sqrt{1+2e^x} \cdot \sqrt{\frac{(1+e^x)^2}{1+2e^x}} dx = \int_0^1 2\pi(1+e^x) dx = 2\pi(x+e^x) \Big|_0^1$$

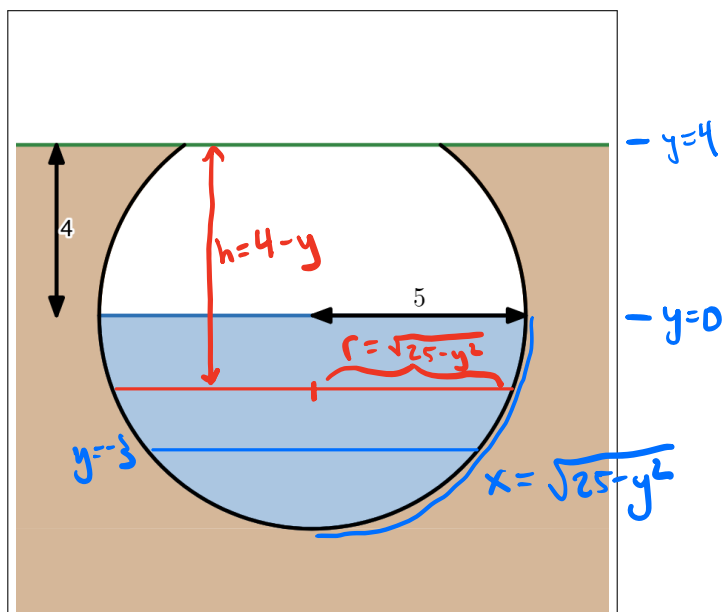
$$= 2\pi(1+e-1)$$

$$= \boxed{2\pi e}$$

B4 - Applications of integration to physics and engineering (inspired by part 1 of the class notes and Brightspace problem 3)

- (a) A tank is the shape of a sphere of radius 5 meters with the top 1 meter cut off. This tank is buried underground so that the top of the tank is right at the surface, and the center of the sphere is 4 meters below the surface (shown to the right). If the tank is half full of a liquid of density ρ , setup an integral representing the work required to pump water to the surface until the depth of the water in the tank is 2 meters. You do not need to evaluate the integral. Use g for gravity.

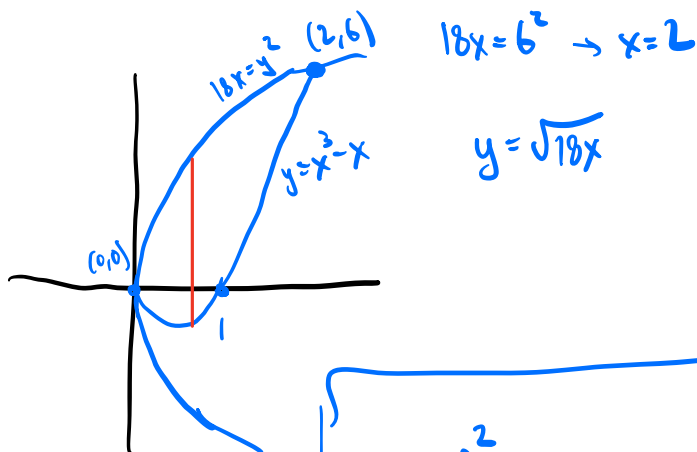
$$W = \int_{-3}^0 g \cdot \rho \cdot (4-y) \cdot \pi (\sqrt{25-y^2})^2 dy$$



- (b) Consider the lamina in the plane of density $\rho = 1$ bounded by the curves $y = x^3 - x$ and $18x = y^2$. These curves intersect at two points where $y = 0$ and $y = 6$. Find each of the following. You can express your answers using unsimplified and unevaluated integrals.

- (i) The moment of the lamina about the y -axis

- (ii) The y -coordinate of the center of mass



$$\bar{y} = \frac{M_x}{m} = \frac{\int_0^2 \frac{1}{2} ((\sqrt{18x})^2 - (x^3 - x)^2) dx}{\int_0^2 (\sqrt{18x} - (x^3 - x)) dx}$$

$$M_y = \int_0^2 x (\sqrt{18x} - (x^3 - x)) dx$$

B5 - Areas and lengths in polar coordinates (inspired by past final worksheet problem 2 and Brightspace problem 10)

Consider the polar curve $r = 1 + 2 \cos \theta$.

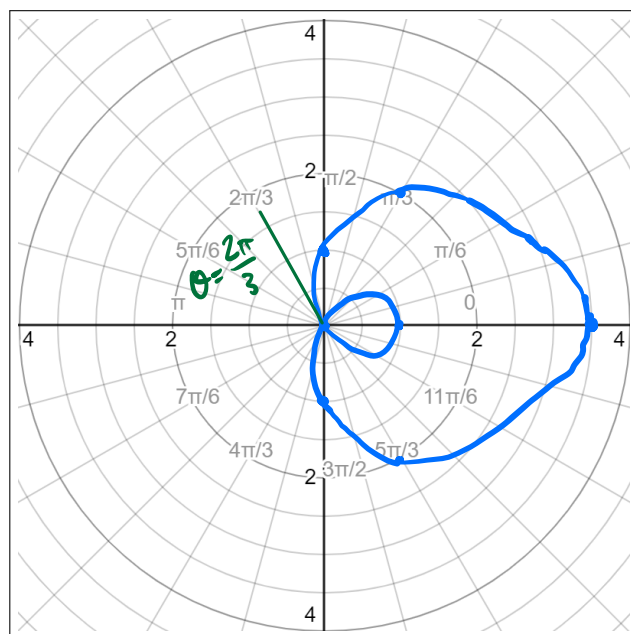
- (a) Sketch the polar curve on the grid to the right. Clearly mark any intercepts with the x and y -axes.

θ	r
0	3
$\pi/2$	1
$2\pi/3$	0
$3\pi/2$	-1
$5\pi/2$	0
3π	-1

$$1 + 2 \cos \theta = 0$$

$$\cos \theta = -\frac{1}{2}$$

$$\theta = \frac{2\pi}{3}, \frac{4\pi}{3}$$



- (b) Setup (but don't evaluate) an integral representing each of the following:

- (i) The total length of the polar curve

$$r' = (-2 \sin \theta)$$

$$L = \int_0^{2\pi} \sqrt{(1 + 2 \cos \theta)^2 + (-2 \sin \theta)^2} d\theta$$

- (ii) The area bounded by the polar curve in the second quadrant (i.e. $x \leq 0$ and $y \geq 0$)

$$A = \int_{\pi/2}^{2\pi/3} \frac{1}{2} (1 + 2 \cos \theta)^2 d\theta$$